

Course Title	Quantum Algorithms for Applications							
Course Code	CSHO533QCP							
						Category	Informal Lab	
		L	T	P			HONOURS	NO
Contact Hrs./Week		4	0	2	CIA Marks		50	
Contact Hrs./Sem.		60	0	30	ESE Marks		50	
Credits.		4	0	5	ESE Exam Hours		3	
Course Description: In this course, students will build upon the basic concepts learned in the “Introduction to Quantum Computing” course and in “Advanced Quantum Computing” course. The course covers the new generation application based hybrid quantum-classical algorithms like QAOA, VQE, and also alternative technologies like Quantum Annealing used for Quantum Optimization. The course will involve projects in advanced applications using Python programming.								
Course Objective: After completing the course, students should be able to – • Apply quantum optimization techniques to solve complex optimization problems • Analyze the efficiency of quantum optimization algorithms compared to classical methods • Implement QML algorithms to enhance machine learning tasks using quantum computing • Design and analyze QNN architectures for improved performance in pattern recognition and data analysis • Implement QAOA to approximate solutions for combinatorial optimization problems • Evaluate the performance and applications of QAOA in solving practical optimization tasks • Apply VQE for finding ground state energies of quantum systems • Integrate VQE techniques with classical optimization methods to solve complex problems • Explore the principles and applications of quantum annealing in optimization • Implement quantum annealing techniques to solve real-world problems								
Prerequisites: Advanced Quantum Computing, Introduction to Quantum Computing								
Syllabus						Teaching Hours (60T+30P)		
Unit 1 : Quantum Optimization Quantum optimization fundamentals, Combinatorial optimization, quantum vs. classical optimization methods, quantum annealing, problem encoding, Hamiltonian formulation, performance analysis, Real-world applications in logistics, finance, and resource allocation. Practical Experiments: Implementation of quantum optimization algorithms.						12T+6P		
Text Books: T1. "Quantum Computation and Quantum Information" by M. A. Nielsen and I. L. Chuang. Reference Books: R1. "Quantum Computing: An Applied Approach” by Jack Hidary. Self-Study: -								
Topics Contributing to Needs of								
Local: -		Employability: Quantum Annealing Algorithm			Professional Ethics: -			
Regional: -		Skill Devops: Quantum Annealing Algorithm			Human Values: -			
National: Quantum Annealing Algorithm		Entrepreneurship: -			Environment & Sustainability: -			
Global: Quantum Annealing Algorithm					Gender Issues: -			

Unit 2: Quantum Machine Learning (QML) and Quantum Neural Networks (QNNs) Quantum data encoding, quantum, feature spaces, quantum kernels, quantum support vector machines (QSVM), quantum principal component analysis (QPCA), quantum neural network(QNN) architectures, training quantum models, hybrid quantum-classical algorithms, applications in classification, clustering, and regression tasks Practical Experiments: Implementation of QML and QNN algorithms			12T+6P
Text Books: T1. "Quantum Computation and Quantum Information" by M. A. Nielsen and I. L. Chuang. Reference Books: R1. "Quantum Computing: An Applied Approach" by Jack Hidary. Self-Study: -			
Topics Contributing to Needs of			
Local: -	Employability: QSVM, QNN, QPCA	Professional Ethics: QSVM, QNN, QPCA	
Regional: -	Skill Devops: QSVM, QNN, QPCA	Human Values: -	
National: QSVM, QNN, QPCA	Entrepreneurship: -	Environment & Sustainability: -	
Global: QSVM, QNN, QPCA		Gender Issues: -	
Unit 3: Quantum Approximate Optimization Algorithm (QAOA) QAOA fundamentals, problem formulation, Hamiltonian mixers, cost Hamiltonians, variational parameters, optimization landscapes, classical optimizers, depth vs. performance trade-offs, benchmarking QAOA, applications in scheduling, portfolio optimization and network design Practical Experiments: Implementation of QAOA algorithms			12T+6P
Text Books: T1. "Quantum Computation and Quantum Information" by M. A. Nielsen and I. L. Chuang. Reference Books: R1. "Quantum Computing: An Applied Approach" by Jack Hidary. Self-Study: -			
Topics Contributing to Needs of			
Local: -	Employability: QAOA algorithms	Professional Ethics: QAOA algorithms	
Regional: -	Skill Devops: QAOA algorithms	Human Values: -	
National: QAOA algorithms	Entrepreneurship: -	Environment & Sustainability: -	
Global: QAOA algorithms		Gender Issues: -	
Unit 4: Variational Quantum Eigensolver (VQE) VQE principles, ansatz selection, parameterized quantum circuits, energy minimization, classical optimization techniques, hybrid quantum-classical workflows, applications in quantum chemistry, finding ground state energies, material science problems, benchmarking VQE performance Practical Experiments: Implementation of VQE algorithm			12T+6P
Text Books: T1. "Quantum Computation and Quantum Information" by M. A. Nielsen and I. L. Chuang. Reference Books: R1. "Quantum Computing: An Applied Approach" by Jack Hidary. Self-Study: -			
Topics Contributing to Needs of			

Local: -	Employability: VQE algorithms	Professional Ethics: VQE algorithms														
Regional: -	Skill Devops: VQE algorithms	Human Values: -														
National: VQE algorithms	Entrepreneurship: -	Environment & Sustainability: -														
Global: VQE algorithms		Gender Issues: -														
Unit 5: Quantum Annealing for Optimization: Case Study Quadratic Unconstrained Binary Optimization (QUBO), Define a QUBO problem, Submit to the annealer (either neal or D-Wave), Receive and interpret results,D-Wave Systems: A Quantum Annealer. Practical Experiments: Mini-Project		12T+6P														
Text Books: T1. "Quantum Computation and Quantum Information" by M. A. Nielsen and I. L. Chuang. Reference Books: R1. "Quantum Computing: An Applied Approach” by Jack Hidary. Self-Study: -																
Topics Contributing to Needs of																
Local: -	Employability: QUBO	Professional Ethics: QUBO														
Regional: -	Skill Devops: QUBO	Human Values: -														
National: QUBO	Entrepreneurship: -	Environment & Sustainability: -														
Global: QUBO		Gender Issues: -														
Course Outcomes (CO)		Knowledge Level	Units Mapping													
CO1: Apply quantum optimization techniques to solve complex optimization problems and analyze their efficiency compared to classical methods.		L3	U1													
CO2: Implement QML algorithms and design QNN architectures for enhanced machine learning tasks and improved performance in pattern recognition and data analysis.		L3	U2													
CO3: Implement the Quantum Approximate Optimization Algorithm (QAOA) to approximate solutions for combinatorial optimization problems and evaluate its performance.		L3	U3													
CO4: Apply the Variational Quantum Eigensolver (VQE) for finding ground state energies of quantum systems and integrate VQE with classical optimization methods.		L3	U4													
CO5: Apply the principles and applications of quantum annealing in optimization to implement quantum annealing techniques and solve real-world problems.		L3	U5													
Mapping with Program Outcomes (CO - PO/PSO)																
CO	PO												PSO			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	3	2	2	1	3	1	1				1					
CO2	3	3	2	1	3	2	1				1					
CO3	3	3	2	2	3	1	1	1		2	1					

CO4	3	3	2	2	3	1	1	1	1	2	1					
CO5	3	3	2	2	3	2	1	1	1	2	1					
Course Designed: Dr Diana Jeba Jingle I																
Course Reviewed: Dr. Debarka																
Revision as on Dated:			23-10-2025													